ISM Companion Reader Session 10

Student's t-Test, Chi-Square & Maximum Likelihood Estimation

Small-sample testing, contingency independence, goodness-of-fit, and MLE point estimation.

1 Student's t-Distribution for Small Samples

When sample size is small ($n < 30$) and the population standard deviation σ is unknown, we estimate σ using the sample standard deviation S. This extra uncertainty thickens the distribution tails, requiring Student's t-distribution with $df = n - 1$.

t-Test Statistic Formula:

$$t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

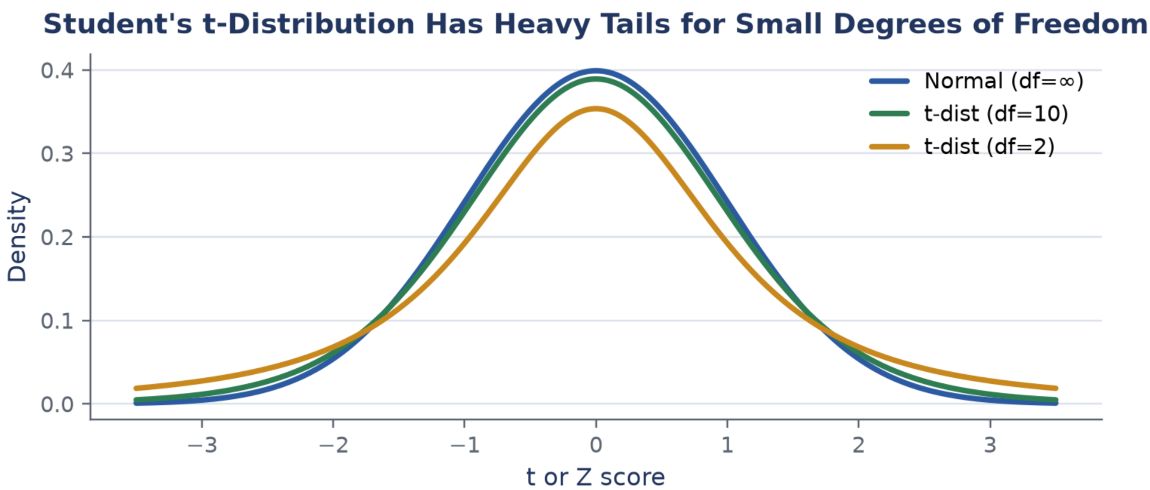

Figure: Student's t-distributions have heavier tails than the standard normal distribution for small degrees of freedom.

Slide Example: Sports Car Annual Mileage

Problem Statement: Sports car owners claim their cars are driven an average of $\bar{X} = 22,500$ km/year with sample standard deviation $S = 3,000$ km based on $n = 10$ cars. Test the hypothesis that $\mu = 20,000$ km vs $H_1 : \mu > 20,000$ km at $\alpha = 0.01$.

Step 1: State Hypotheses: $H_0 : \mu \leq 20,000$ vs $H_1 : \mu > 20,000$.

Step 2: Degrees of Freedom & Critical Value: $df = n - 1 = 9$. For right-tailed t-test at $\alpha = 0.01$, $t_{\text{crit}} = 2.821$.

Step 3: Compute t Test Statistic:

$$t_{\text{cal}} = \frac{22500 - 20000}{3000/\sqrt{10}} = \frac{2500}{948.68} \approx 2.635$$

Step 4: Make Decision: Since $t_{\text{cal}} = 2.635 < 2.821$, we fail to reject H_0. There is insufficient evidence at $\alpha = 0.01$ to prove higher annual mileage.

2 Chi-Square Test (χ^2)

The Chi-Square test evaluates categorical data. It comes in two primary forms:

- **Test of Independence:** Checks if two categorical variables (e.g., smoking habit vs hypertension) are independent in a contingency table.
- **Goodness-of-Fit Test:** Checks if observed sample counts fit a theoretical distribution (e.g., Poisson or Uniform).

Chi-Square Test Statistic:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}, \quad \text{where } E_i = \frac{\text{Row Total} \times \text{Col Total}}{\text{Grand Total}}$$

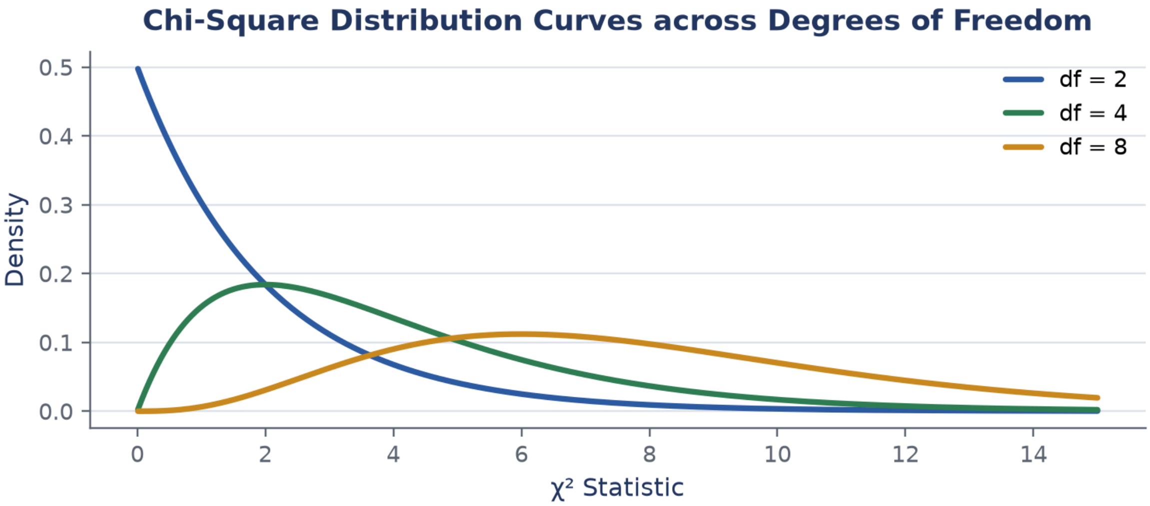

Figure: Chi-Square distribution shapes across degrees of freedom ($df = 2, 4, 8$).

3 Maximum Likelihood Estimation (MLE)

Maximum Likelihood Estimation (MLE) finds the parameter value $\hat{\theta}$ that maximizes the likelihood of observing the collected sample data.

Likelihood Function:

$$L(\theta) = f(x_1;\theta) \cdot f(x_2;\theta) \cdots f(x_n;\theta) = \prod_{i=1}^{n} f(x_i;\theta)$$

In practice, we maximize the **Log-Likelihood** $\ln L(\theta)$ by setting its derivative to zero.

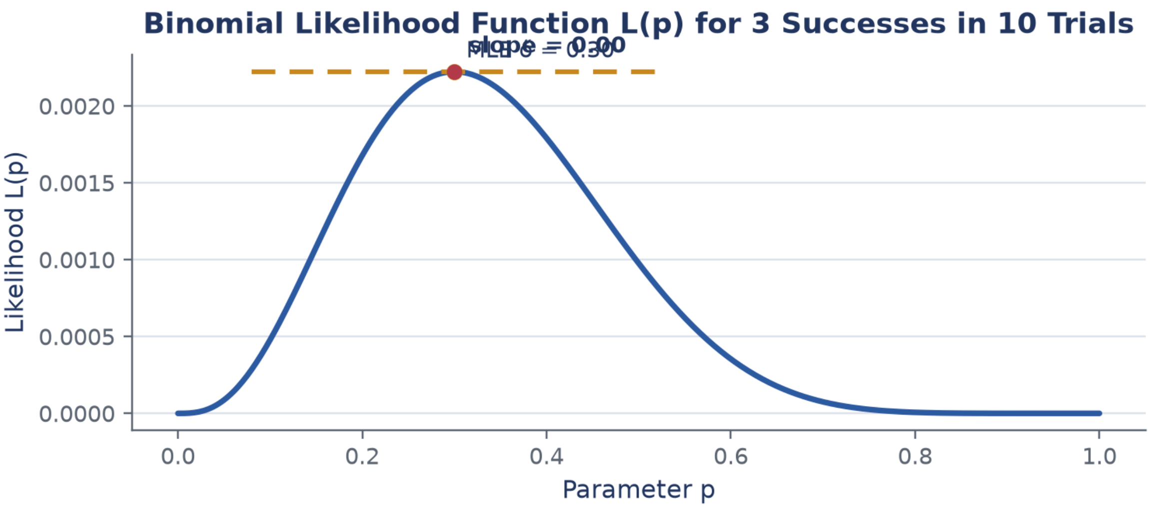

Figure: Likelihood function $L(p)$ for binomial trial yielding 3 successes in 10 flips, peaking at $\hat{p} = 0.30$.

Slide Example: Binomial MLE

Problem Statement: In a quality check of 10 bike helmets, 3 are found flawed ($k = 3, n = 10$). Find the Maximum Likelihood Estimator $\hat{p}$ for the flaw probability p.

Step 1: Write Likelihood Function:

$$L(p) = \binom{10}{3} p^3 (1-p)^7$$

Step 2: Take Log-Likelihood:

$$\ln L(p) = \ln \binom{10}{3} + 3 \ln p + 7 \ln(1-p)$$

Step 3: Differentiate and set to 0:

$$\frac{d}{dp} \ln L(p) = \frac{3}{p} - \frac{7}{1-p} = 0 \implies 3(1-p) = 7p \implies 3 = 10p \implies \hat{p} = \frac{3}{10} = 0.30$$

Summary Checklist

- **t-Test:** Use for $n < 30$ when σ is unknown ($df = n - 1$).
- **Chi-Square (χ^2):** Evaluates categorical independence ($df = (r-1)(c-1)$) and goodness-of-fit ($df = k - 1 - p$).
- **MLE:** Maximizes sample probability by solving $\frac{d}{d\theta} \ln L(\theta) = 0$.